

FEMamba: A Feature-Enhanced Mamba Framework with Degradation-Stage Global Regularization for Bearing Remaining Useful Life Prediction

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Abstract: Accurate prediction of bearing Remaining Useful Life (RUL) is crucial for implementing predictive maintenance and ensuring safe operation of industrial equipment. However, existing methods still face challenges in handling multi-source heterogeneous time series data and modeling long-term degradation processes. This paper proposes a novel Feature Enhanced Mamba framework (FEMamba) that achieves high-precision bearing RUL prediction by integrating a Multi-Feature Enhancement Selection module (MFES) and a Degradation Stage-based Global Regularization strategy (DSGR). Specifically, the MFES module effectively processes heterogeneous sensor data through an adaptive feature selection mechanism, while the DSGR strategy optimizes the physical significance of feature representations by constructing feature space constraints that consider degradation dynamics. Experiments on the PHM2012 dataset demonstrate that our proposed method outperforms existing approaches across all evaluation metrics, achieving an R^2 value of 0.9601 and a Mean Square Error (MSE) of 3.22×10^{-3} . Ablation studies further verify the effectiveness of the MFES module and DSGR strategy, showing MSE reductions of 7.9% and 10.6% respectively compared to the baseline model. These results indicate that our proposed method provides an accurate and reliable solution for bearing RUL prediction.

Key Words: Remaining Useful Life Prediction, Bearing Health Monitoring, Time Series Analysis, State Space Modeling

1 Introduction

Bearing life prediction is a crucial task in industrial equipment predictive maintenance, aiming to accurately estimate the Remaining Useful Life (RUL) of bearings based on historical operational data. In industrial settings, bearings serve as core components of rotating machinery, with their performance directly impacting the reliability and safety of entire equipment systems. The degradation process of bearings typically manifests as a nonlinear, multi-stage dynamic process involving various failure modes. In recent years, with the rapid development of Industry 4.0 and intelligent manufacturing, advances in deep learning technology have provided new solutions for bearing life prediction. Research has shown that deep learning models can automatically extract complex feature representations from vast monitoring data and effectively model bearing degradation processes [1]. In recent years, contrastive learning has demonstrated significant advantages in mechanical health prediction. Ragab et al. [2] proposed a Contrastive Adversarial Domain Adaptation (CADA) method for machine RUL prediction, which demonstrated superior performance in various cross-domain scenarios by incorporating contrastive loss into the adversarial domain adaptation architecture while considering target-specific information. Kong et al. [3] proposed a contrastive learning framework enhanced with unlabeled samples, effectively addressing the challenge of insufficient labeled data. To improve the accuracy and generalization capability of bearing RUL prediction, Zhang et al. [4] developed a supervised contrastive learning-based domain adaptation net-

work that enhanced model performance in cross-domain scenarios by extracting more discriminative feature representations. Recently, Xie et al. [5] proposed an incremental contrastive hybrid model for online RUL prediction, combining contrastive learning transformer (CLformer) with Enhanced Generalized Wiener Process (EGWP) to describe mechanical degradation trends, demonstrating the potential of contrastive learning in addressing RUL prediction uncertainties. Despite significant progress in RUL prediction using contrastive learning, challenges remain in model generalization, computational efficiency, and uncertainty quantification, which are key issues that future research needs to address. However, existing research primarily applies contrastive learning to feature extraction and domain adaptation, lacking utilization of the continuity characteristics inherent in bearing life prediction labels. Unlike discrete classification tasks, bearing RUL prediction is a continuous value regression problem, where label pairs naturally exhibit correlation. Samples with similar remaining lifetimes often share comparable degradation features and state manifestations, while samples with significantly different remaining lifetimes demonstrate characteristics of distinct degradation stages. Therefore, designing appropriate contrastive learning strategies to fully leverage the similarity information embedded in continuous label pairs represents a crucial research direction for improving bearing life prediction performance. Our contributions are as follows:

- We propose a Feature Enhanced Mamba (FEMamba) architecture that effectively processes heterogeneous

time series data in bearing monitoring through a Multi-Feature Enhancement Selection (MFES) module, while leveraging Mamba's selective state space mechanism for efficient long sequence modeling. This architecture innovatively addresses the dual challenges of multi-source feature fusion and long-term dependency modeling.

- We design a Degradation Stage-based Global Regularization (DSGR) strategy that explicitly incorporates the physical correlations of continuous labels into the feature learning process by constructing feature space constraints that consider degradation dynamics. This strategy not only ensures prediction accuracy but also optimizes the physical significance of feature representations.
- We conduct a series of experiments on the PHM2012 dataset [6], demonstrating the competitive performance of our method.

2 Methodology

2.1 Problem Formulation

Bearing RUL prediction is a typical regression problem based on multi-source time series data. In industrial settings, bearing health monitoring typically relies on vibration signals collected from multiple sensors and domain expert-designed time-frequency features. Given a time series sample containing raw vibration signals and manually extracted features:

$$X = \{x_t\}_{t=1}^T \in \mathbb{R}^{T \times C}$$

where T represents the sequence length and C represents the total feature dimension. The feature dimension can be further decomposed into the number of raw vibration signal channels C_r and manually extracted feature dimensions C_h :

$$C = C_r + C_h$$

In the RUL prediction task, our goal is to predict the normalized remaining life value through modeling of the input sequence. To ensure good numerical properties of the prediction target, we normalize the actual Remaining Useful Life (RUL) by the maximum lifetime value:

$$y = \frac{RUL}{RUL_{max}} \in [0, 1]$$

This normalization strategy not only aids model training stability but also provides the prediction results with clear physical meaning and boundary constraints.

However, bearing RUL prediction tasks face significant technical challenges. First, the heterogeneity of input features presents difficulties in feature fusion. Raw vibration signals contain rich temporal information, while manually extracted features embody expert knowledge and frequency domain characteristics. How to effectively fuse these features of different scales and physical meanings is a key issue. Second, bearing degradation is a complex long-term cumulative process, requiring models to capture long-sequence temporal dependencies. Additionally, as RUL is a continuously changing physical quantity, predicted values at adjacent time points should maintain sufficient smoothness and continuity.

Based on these considerations, we formalize the RUL prediction task as an end-to-end prediction problem. Denoting the parameterized prediction model as f_θ , the prediction process can be expressed as:

$$\hat{y} = f_\theta(X)$$

where $\hat{y}$ is the predicted normalized RUL value. To ensure both prediction accuracy and feature representation effectiveness, we design a composite loss function:

$$\min_{\theta} \mathcal{L}(y, \hat{y}, h)$$

where h represents the model's intermediate feature representation. This design not only focuses on prediction accuracy but also enhances model generalization ability through feature-level constraints.

In the following sections, we will detail the prediction model architecture based on the Feature-Enhanced Mamba Framework and the design of Degradation-Stage Global Regularization to effectively address these technical challenges.

2.2 Model Framework

Before establishing the prediction model, we first perform feature extraction and preprocessing on the raw vibration signal $s(t) \in \mathbb{R}^T$. Feature extraction includes statistical features (mean, variance, skewness), waveform features (peak factor, impulse factor, margin factor), frequency domain features (spectrum line shape indicators, energy ratios), and complexity indicators (kurtosis, entropy, fractal dimension). These features, together with the raw vibration signals, constitute the model's input sequence $X = \{x_t\}_{t=1}^T \in \mathbb{R}^{T \times C}$.

Based on the preprocessed data, we propose an end-to-end prediction framework comprising two core components: the Feature Enhanced Mamba (FEMamba) architecture and the Degradation Stage-based Global Regularization (DSGR) strategy. The FEMamba module first performs fusion and selection of multi-source features, outputs enhanced feature representations through the Mamba module for sequence modeling, and finally obtains the RUL prediction value $\hat{y}$ through fully connected layers, as shown in Fig1. The entire model is optimized through a multi-objective loss function combining prediction error and DSGR regularization:

$$\mathcal{L} = \mathcal{L}_{MSE}(y, \hat{y}) + \lambda \mathcal{L}_{DSGR}(H)$$

This design ensures both prediction accuracy and enhances model generalization ability and physical interpretability through feature space structural constraints. The specific components of the model will be detailed in subsequent sections.

2.3 FEMamba Model Architecture

To effectively handle multi-source heterogeneous time series data in bearing RUL prediction, we propose a Feature Enhanced Mamba (FEMamba) prediction model. As shown in Figure 1, FEMamba consists of two core modules: the Multi-Feature Enhancement and Selection (MFES) module and the Mamba-based sequence modeling module.

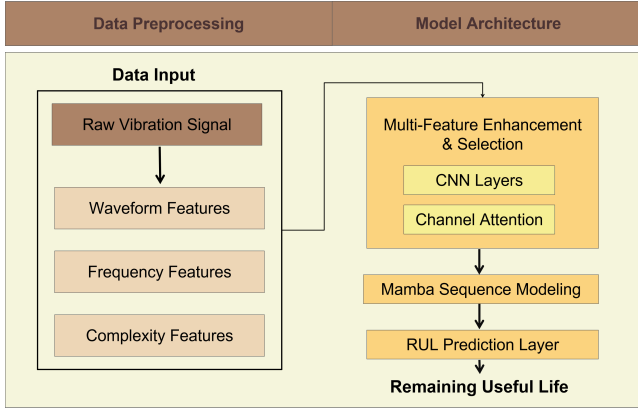


Fig. 1: FEMamba Framework

2.3.1 Multi-Feature Enhancement Selection Module

To address the heterogeneity of raw vibration signals and manually extracted features in bearing monitoring data, we design a two-stage feature enhancement and selection module. This module first performs feature extraction through multi-layer CNN, then uses attention mechanisms for feature selection and reorganization. Specifically, given input sequence X , the feature enhancement process can be expressed as:

$$F_{cnn} = \text{CNN}(X) \in \mathbb{R}^{T' \times C'}$$

where T' and C' represent the sequence length and number of channels after CNN processing. To adaptively select and fuse important features, we design a channel attention-based feature selection mechanism:

$$A = \text{Softmax}(\text{Conv1d}(F_{cnn})) \in \mathbb{R}^{T' \times C'}$$

$$F_{att} = F_{cnn} \odot A$$

where A is the channel attention weight matrix, and $\odot$ represents element-wise multiplication. Finally, the enhanced features are projected into a unified feature space through a feature mapping layer:

$$H = \text{Conv1d}(F_{att}) \in \mathbb{R}^{T' \times D}$$

This design not only effectively extracts and fuses multi-source features but also performs adaptive selection based on feature importance, improving the model's perception of key features.

2.3.2 Mamba Sequence Modeling Module

Considering the long-term dependencies in bearing degradation processes, we adopt the Mamba structure for sequence modeling. Compared to traditional Transformers, Mamba achieves more efficient long sequence modeling through the Selective State Space (S4) mechanism [7]. Given the enhanced feature sequence H , the computation process of the Mamba module is:

$$Y = \text{Mamba}(H) = \text{S4}(H, \Theta)$$

where Θ represents the learnable parameters of the Mamba module. Specifically, the S4 mechanism implements

sequence modeling through the following state update equations:

$$x_t = \text{SiLU}(W_i x_t + b_i)$$

$$s_t = A s_{t-1} + B x_t$$

$$y_t = C s_t + D x_t$$

where A, B, C, D are learnable state space parameters, and s_t is the hidden state. Finally, the prediction result is obtained through global average pooling and fully connected layers:

$$\hat{y} = \sigma(\text{FC}(\text{AvgPool}(Y)))$$

This design enables the model to efficiently capture long-term dependencies in sequences while maintaining low computational complexity. Through the organic combination of MFES and Mamba modules, FEMamba effectively addresses both feature selection and long-term dependency modeling challenges, providing an efficient and effective solution for bearing RUL prediction.

2.4 Degradation Stage-based Global Regularization Strategy

Bearing degradation is a continuous physical process, and samples from similar degradation stages should have similar representations in feature space. Based on this understanding, we propose the Degradation Stage-aware Global Regularization (DSGR) strategy. This strategy optimizes the physical meaning of feature representations while ensuring prediction accuracy through feature space constraints that consider degradation dynamics.

The core of DSGR is a dual-objective optimization framework that integrates prediction accuracy and feature space structure. The overall loss function is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{base}} + \lambda \mathcal{L}_{\text{reg}}$$

where the base loss $\mathcal{L}_{\text{base}}$ uses mean squared error to ensure prediction accuracy:

$$\mathcal{L}_{\text{base}} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

The global regularization loss $\mathcal{L}_{\text{reg}}$ implements feature space structure optimization through:

$$\mathcal{L}_{\text{reg}} = \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N \left(1 - \exp \left(-\frac{d(f_i, f_j)}{\tau} \right) \right) \cdot |y_i - y_j|$$

Here, $d(f_i, f_j)$ represents the Euclidean distance between sample pairs in feature space, τ is a temperature parameter, and $|y_i - y_j|$ measures the difference in normalized remaining life. Through the explicit introduction of degradation stage difference $|y_i - y_j|$, consistency between feature space structure and actual physical degradation processes is ensured. Meanwhile, the exponential form $\exp(-d/\tau)$ provides smooth gradient flow, and the temperature parameter τ can flexibly adjust the sensitivity of distance measurement.

From a theoretical perspective, DSGR achieves efficient optimization of feature space through a concise mathematical form. The convexity of the base loss term provides a stable foundation for optimization, while consideration of global sample pairs ensures complete topological structure of the feature space.

3 Experimental Details

3.1 Experimental Setup

This study utilizes the IEEE PHM 2012 Challenge dataset for experimental validation [6]. For data selection, we use data from bearings 1-1 and 1-2 to construct the training set, with bearing 1-3 data serving as the test set, focusing on analyzing horizontal vibration signals. For feature extraction, we construct a comprehensive feature set covering statistical features (mean, variance, skewness), waveform features (peak factor, impulse factor, margin factor), frequency domain features (spectrum line shape indicators, energy ratios), and complexity indicators (kurtosis, entropy, fractal dimension). To enhance the model's ability to learn temporal features, we segment the original signal sequences into subsequences of length 20, a processing strategy that both preserves local temporal dependencies and effectively expands the training sample size.

For model training, we adopt a mini-batch training strategy with a batch size of 32, using Adam as the optimizer with a learning rate of 0.0003. For the model's core components, we set both the DSGR regularization parameter λ and temperature coefficient τ to 0.1, with the Mamba block's hidden dimension set to 128. These hyperparameter choices are based on preliminary experimental results, achieving a good balance between model performance and computational efficiency.

All experiments were conducted on a server equipped with an NVIDIA RTX 4090D (24GB VRAM) GPU and Intel(R) Xeon(R) Platinum 8474C processor (15 vCPU). The software environment uses Ubuntu 20.04 operating system with Python 3.8 and PyTorch 1.10.0 deep learning framework, CUDA version 11.3.

3.2 Evaluation Metrics

To comprehensively evaluate the model's prediction performance, we employ multiple complementary evaluation metrics. First, we use the coefficient of determination (R^2) to measure the model's ability to explain data variability:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

where y_i represents the true value, $\hat{y}_i$ the predicted value, and $\bar{y}$ the mean of true values. R^2 ranges from [0,1], with values closer to 1 indicating predictions closer to true values.

For error evaluation, we use Mean Squared Error (MSE) and its square root (RMSE) to measure the deviation between predicted and true values:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

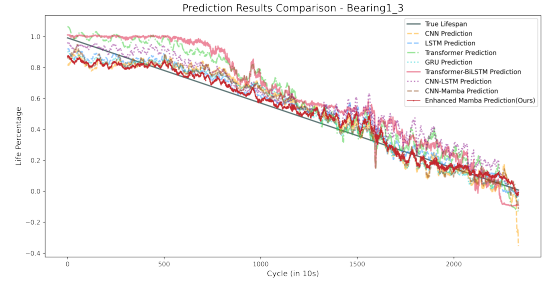


Fig. 2: Comparison of the performance of different models

Additionally, we introduce Mean Absolute Error (MAE) to provide an intuitive error measure:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

3.3 Experimental Results Analysis

As shown in Table 1 and Fig 2, detailed analysis of the comparative experimental results shows that our proposed FEMamba+DSGR model achieves optimal performance across all evaluation metrics. Specifically, the model achieves an R^2 value of 0.9601, with MSE reduced to 3.22×10^{-3} , and RMSE and MAE reaching 5.68×10^{-2} and 4.55×10^{-2} respectively, demonstrating significant advantages over other baseline models.

Table 1: Performance Comparison of Different Models

Model	R^2	MSE(10^{-3})	RMSE(10^{-2})	MAE(10^{-2})
GRU	0.9537	3.73	6.11	5.18
LSTM	0.9523	3.85	6.20	5.22
Transformer	0.8194	14.56	12.07	10.29
Transformer-BiLSTM	0.7369	21.21	14.56	13.07
CNN	0.9295	5.68	7.54	6.41
CNN-LSTM	0.8640	10.96	10.47	9.06
CNN	0.7636	19.05	13.81	12.16
-Transformer				
Ours	0.9601	3.22	5.68	4.55

From a performance comparison perspective, while basic recurrent neural network models (GRU and LSTM) perform well with R^2 values of 0.9537 and 0.9523 respectively, they still fall short of our proposed model. The Transformer-based model series (including pure Transformer, Transformer-BiLSTM, and CNN-Transformer) show relatively weaker performance on this task, with R^2 values not exceeding 0.82, possibly due to their limitations in handling continuous value regression tasks.

Notably, our FEMamba+DSGR model achieves optimal performance across all metrics: the R^2 value improves by 0.0064 compared to the second-best model (GRU), demonstrating stronger fitting capability; MSE decreases by 13.7% compared to GRU (from 3.73×10^{-3} to 3.22×10^{-3}), while RMSE and MAE decrease by 7.0% and 12.2% respectively.

These experimental results comprehensively validate the effectiveness of our proposed Feature Enhanced Mamba model architecture and Degradation Stage Global Regularization strategy, demonstrating significant performance ad-

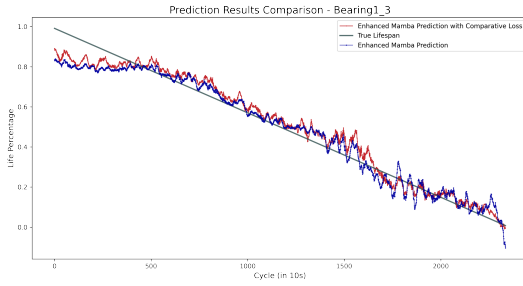


Fig. 3: The comparison before and after using the DSGR strategy

vantages in bearing RUL prediction tasks. The model not only effectively handles multi-source heterogeneous time series data but also accurately captures long-term dependencies in bearing degradation processes.

3.4 Ablation Study Analysis

To thoroughly validate the effectiveness of our proposed method, we designed two groups of ablation experiments to verify the contributions of the Multi-Feature Enhancement Selection (MFES) module and Degradation Stage-based Global Regularization (DSGR) strategy.

First, we evaluated the effectiveness of the MFES module. As shown in Table 2, compared to the simple CNN+Mamba structure, FEMamba with the MFES module shows significant improvements across all metrics. Specifically, the R^2 value increases from 0.9515 to 0.9553, MSE decreases from 3.91×10^{-3} to 3.60×10^{-3} , and RMSE decreases from 6.25×10^{-2} to 6.00×10^{-2} . This improvement clearly demonstrates the effectiveness of the MFES module in handling heterogeneous time series features, validating its crucial role in feature selection and fusion processes.

Table 2: Ablation Study of MFES Module

Model	R^2	MSE ($\times 10^{-3}$)	RMSE ($\times 10^{-2}$)	MAE ($\times 10^{-2}$)
CNN+Mamba	0.9515	3.91	6.25	5.26
FEMamba	0.9553	3.60	6.00	4.45

Second, we verified the importance of the DSGR strategy. As shown in Table 3 and Fig 3, after introducing DSGR to FEMamba, model performance further improves. The model's R^2 value increases from 0.9553 to 0.9601, MSE decreases by 10.6% (from 3.60×10^{-3} to 3.22×10^{-3}), and RMSE correspondingly decreases by 5.3% (from 6.00×10^{-2} to 5.68×10^{-2}). These improvements clearly demonstrate that the DSGR strategy effectively enhances prediction accuracy by introducing feature space constraints that consider degradation dynamics.

Table 3: Ablation Study of DSGR Module

Model	R^2	MSE ($\times 10^{-3}$)	RMSE ($\times 10^{-2}$)	MAE ($\times 10^{-2}$)
Without DSGR	0.9553	3.60	6.00	4.45
With DSGR	0.9601	3.22	5.68	4.55

In summary, the ablation study results strongly confirm the effectiveness of our two core innovations. The MFES

module enhances the model's ability to process multi-source features, while the DSGR strategy further optimizes model performance by introducing physical constraints. The combination of these two components makes FEMamba+DSGR an efficient solution for bearing RUL prediction.

4 Conclusion

This paper proposes a Feature Enhanced Mamba framework for bearing Remaining Useful Life prediction, achieving precise modeling of bearing degradation processes through innovative combination of multi-feature selection enhancement and degradation stage global regularization. The main research achievements can be summarized as follows:

The design of the Feature Enhanced Mamba (FEMamba) model effectively addresses the challenge of processing multi-source heterogeneous time series data. Specifically, the Multi-Feature Enhancement Selection (MFES) module significantly improves the model's perception of key features through an adaptive feature selection mechanism, increasing the R^2 value from 0.9515 to 0.9553. The Mamba-based sequence modeling module achieves efficient long sequence modeling through selective state space mechanisms, providing a solid foundation for bearing degradation process analysis.

The proposed Degradation Stage Global Regularization (DSGR) strategy successfully incorporates physical insights into the feature learning process. By constructing feature space constraints that consider degradation dynamics, this strategy not only optimizes the physical significance of feature representations but also further enhances model performance, reducing MSE from 3.60×10^{-3} to 3.22×10^{-3} . This approach of combining physical knowledge with deep learning provides new research directions for industrial equipment health monitoring.

Systematic experimental validation on the PHM2012 dataset demonstrates the effectiveness and superiority of our proposed method. Compared to existing methods, the FEMamba+DSGR framework achieves optimal performance across all evaluation metrics, particularly showing a 0.0064 increase in R^2 value and a 13.7% decrease in MSE compared to the second-best GRU model. These results comprehensively validate the practical value of this method in bearing RUL prediction tasks.

Future work will focus on several aspects: First, exploring the extension of the proposed method to more complex operating conditions and additional types of rotating machinery components; second, investigating the incorporation of uncertainty quantification into the prediction framework to provide more reliable decision support; finally, further optimizing the model structure to reduce computational complexity and meet real-time monitoring requirements. These improvements will help advance the practical application of predictive maintenance technology in industrial settings.

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